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Assignment 1

1 Part 1 — AN Representation

Problem 1

Consider a single AN used for classification in a 2D space with an augmented vector as discussed in class. That is, this AN has a constant input of 1 as its $0th$ input (which is multiplied by w_0) as well as inputs x_1 and x_2 which are multiplied by w_1 and w_2 , respectively. This AN is a summation unit (SU) and its activation function f is a step function:

$$f(net) = \begin{cases} 1, & net \geq 0 \\ 0, & net < 0 \end{cases}$$

Given the weights $w_0 = 0.5$, $w_1 = 1.0$, $w_2 = -0.3$, **draw** this AN.

Solution

Here is a diagram of an artificial neuron:

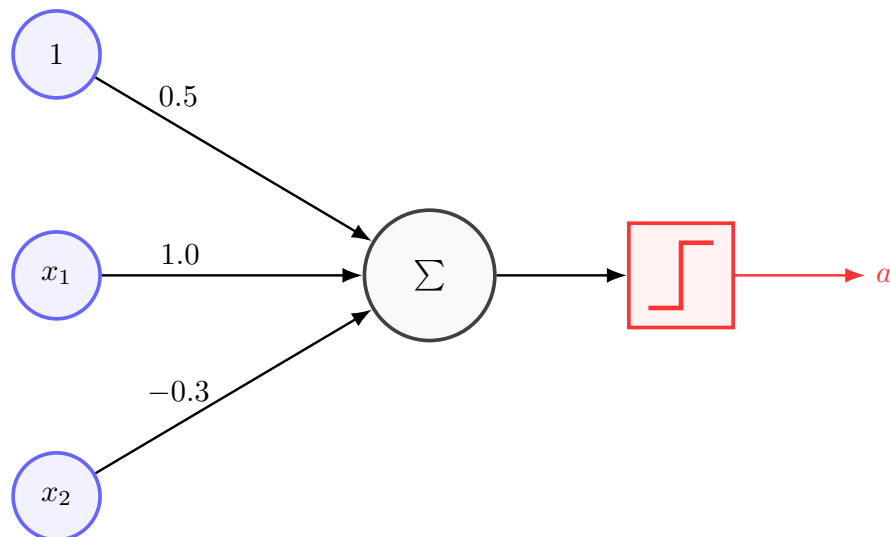


Figure 1: Artificial Neuron (AN) Diagram

Problem 2

Draw (on graph or engineering paper or by using software) the decision boundary encoded by this AN. Be sure to indicate the “font” (Class 1) side of the boundary.

Solution

To identify the slope, we start from the general form of an N -dimensional hyperplane:

$$\begin{aligned}
 &w_0x_0 + w_1x_1 + w_2x_2 = 0 \\
 \Rightarrow &0.5 \cdot 1 + 1.0 \cdot x_1 + (-0.3) \cdot x_2 = 0 \\
 \Rightarrow &0.5 + x_1 - 0.3x_2 = 0 \\
 \Rightarrow &0.5 + x_1 = 0.3x_2 \\
 \Rightarrow &x_2 = \frac{x_1}{0.3} + \frac{0.5}{0.3}
 \end{aligned}$$

This can be plotted as follows:

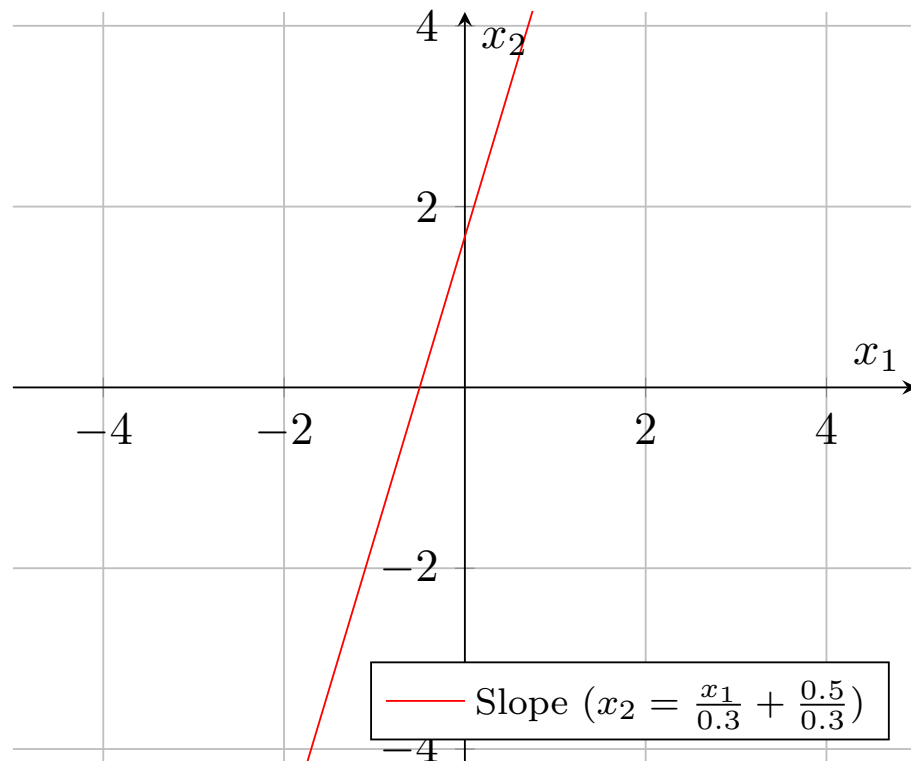


Figure 2: Decision Boundary for the Artificial Neural Unit

To identify the classification side, we choose a test point on one side of the boundary and evaluate the *net* (step) function. For example, take $(0,0)$ as a random test point. Substituting into the hyperplane equation gives:

$$\begin{aligned} 0.5 + x_1 - 0.3x_2 &= 0.5 + 0 - 0.3 \cdot 0 \\ \Rightarrow &= 0.5 \end{aligned}$$

Since $net = 0.5 \geq 0$, the point $(0,0)$ is classified as Class 1. Therefore, the Class 1 (“font”) side of the decision boundary is the half-plane where

$$0.5 + x_1 - 0.3x_2 \geq 0 \iff x_2 \leq \frac{x_1}{0.3} + \frac{0.5}{0.3}.$$

A plot of the boundary with the Class 1 side indicated is shown below:

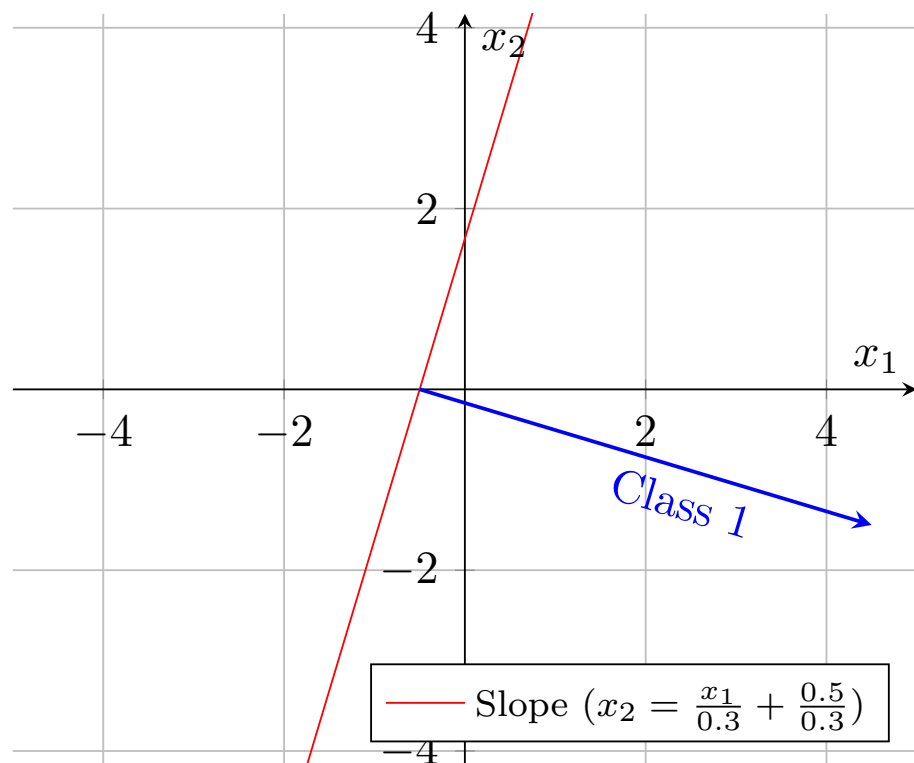


Figure 3: Decision boundary with the Class 1 side indicated

Problem 3

Add the following points on the graph you just drew and **label** the class of each according to the AN.

- (a) $(1.0; 0.2)$
- (b) $(0.0; 0.0)$
- (c) $(-0.1; -0.5)$
- (d) $(1.7; 0.1)$

(e) $(1.8; -1.4)$ Solution

Before adding the points to the graph, let's identify the class of each point. Our equation is $net = 0.5 + x_1 - 0.3x_2$.

(a) Point $(1.0; 0.2)$

$$\begin{aligned} \Rightarrow 0.5 + 1.0 - 0.3 \cdot 0.2 &= 0.5 + 1.0 - 0.06 \\ \Rightarrow &= 1.44 \end{aligned}$$

Since $net = 1.44 \geq 0$, the **point $(1.0; 0.2)$ is classified as Class 1.**

(b) Point $(0.0; 0.0)$

From Problem 2, we already found that the **point $(0.0; 0.0)$ is classified as Class 1.**

(c) Point $(-0.1; -0.5)$

$$\begin{aligned} \Rightarrow 0.5 - 0.1 - 0.3 \cdot (-0.5) &= 0.5 - 0.1 + 0.15 \\ \Rightarrow &= 0.55 \end{aligned}$$

Since $net = 0.55 \geq 0$, the **point $(-0.1; -0.5)$ is classified as Class 1.**

(d) Point $(1.7; 0.1)$

$$\begin{aligned} \Rightarrow 0.5 + 1.7 - 0.3 \cdot 0.1 &= 0.5 + 1.7 - 0.03 \\ \Rightarrow &= 2.17 \end{aligned}$$

Since $net = 2.17 \geq 0$, the **point $(1.7; 0.1)$ is classified as Class 1.**

(e) Point $(1.8; -1.4)$

$$\begin{aligned} \Rightarrow 0.5 + 1.8 - 0.3 \cdot (-1.4) &= 0.5 + 1.8 + 0.42 \\ \Rightarrow &= 2.72 \end{aligned}$$

Since $net = 2.72 \geq 0$, the **point $(1.8; -1.4)$ is classified as Class 1.**

Here is the updated graph:

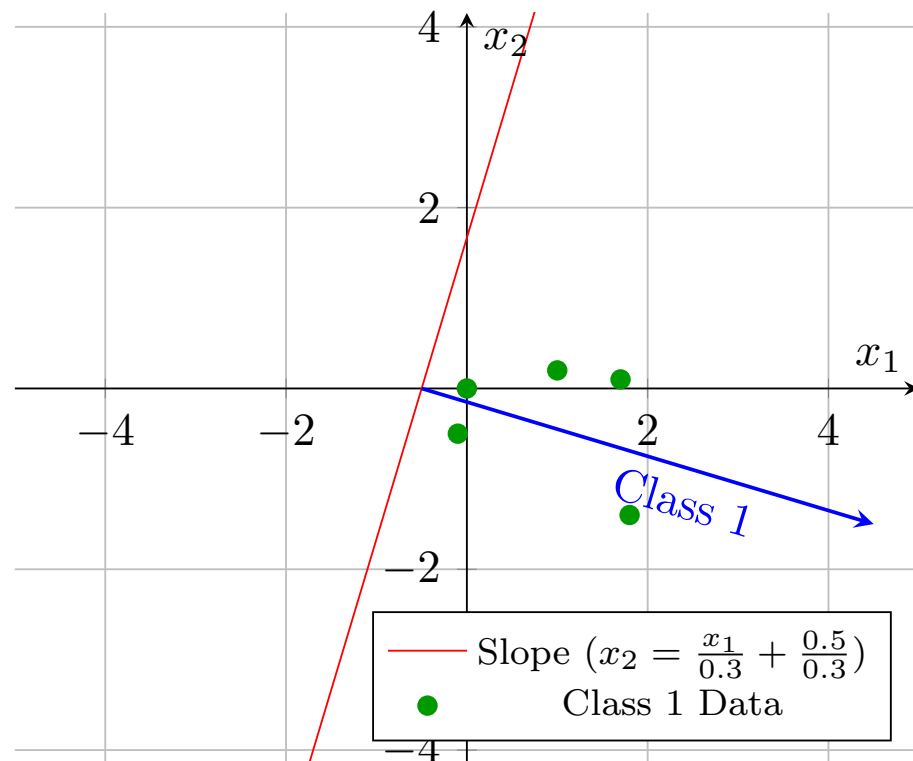


Figure 4: Decision boundary with the Class 1 side indicated

Problem 3

Assume that the AN's classification of each of the points above is correct. **List** a new labeled data item which, if added to the data set above, would necessarily cause the AN to misclassify at least one data item. (That is, given this new data item, there would be no set of possible weights that would allow this AN to correctly classify all data items.) **Explain** why the AN would necessarily misclassify at least one data item.

Solution

After inspecting the plotted data points, we can see that the existing points form a cluster of **Class 1** items. To solve the problem, we need to introduce a new point that breaks this pattern, making it impossible to draw a single straight line that separates two classes. A reliable method is to choose a point that lies on the line segment between two existing points. I selected the existing points $(0.0, 0.0)$ and $(1.0, 0.2)$. If we place a new **Class 0** point at their midpoint, $(0.5, 0.1)$, we create a conflict. This arrangement results in two Class 1 points flanking a Class 0 point on the same line. No single decision boundary can isolate the middle point from the outer two without misclassifying at least one of them.

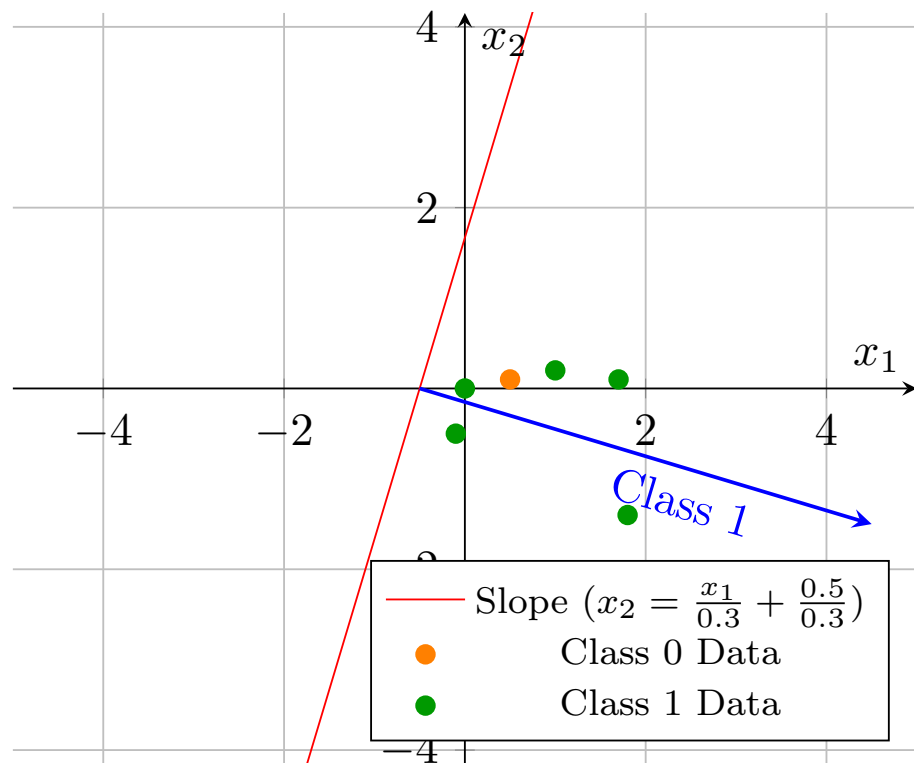


Figure 5: Decision boundary with the Class 1 cluster and an outlier within it

2 Part 2 — AN Learning